

Course Topics

Department: Computing	Program: Bachelor of Information Systems
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First Year Courses

Course Code BUST01 Course Name Quran Recitation & Tajweed Skills	Credit Hours: 1
يعد هذا المقرر من المقررات المهمة؛ كونه متعلق بكلام الله تعالى ثم إنه من متطلبات الجامعة، وهو مقرر نظري وتطبيقي يقوم الطالب فيه بتلاوة وحفظ نصف جزء عم من سورة (الطارق- الناس)، ودراسة الآداب التي ينبغي أن يتحلى بها قارئ القرآن، وفهم وتطبيق هذه الأحكام عند قراءة القرآن الكريم، وتستخدم طريقتي الحوار والمناقشة في التدريس، ويتم تقييم الطالب من خلال تلاوته للقرآن أثناء اللقاءات التعليمية، والقيام بالتكليفات المصاحبة - الصفية والا صفية - مع الامتحانين النصفين والنهائي.	
Course Code BUST01 Course Name English Language (1)	Credit Hours: 4
The current course provides students with the language basics of everyday English to help them communicate in different real-life situations. The course focuses more on real life conversations and the basic grammars that will help each one in his/her major. Moreover, the course provides students with plenty of writing and speaking practices. The updates of the book allow students to be posted with the latest language uses and functions that are easy and useable in real life situations.	
Course Code BUST02 Course Name Arabic Language	Credit Hours: 3
يهتم المقرر بتنمية المهارات اللغوية الأساسية للطالب كونه من متطلبات الجامعة ويتناول جملة من النشاطات الاتصالية والدروس اللغوية والإملائية والتركيبية ويحتوي على معارف ونصوص وتدريب لغوية، ويتألف من كتابين: الأول يركز على مهارتي الاستماع والتحدث، والثاني يركز على مهارتي القراءة والكتابة، مع احتواء كل من الكتابين على أساسيات التركيب النحوي وبعض القواعد الإملائية ويعتمد المقرر أسلوب التدريبات والتطبيقات العملية لكل طالب، بالإضافة إلى النصوص التطبيقية في CD المرفق بالكتاب مع الامتحان النصفين والنهائي.	
Course Code BUST04 Course Name Computer Skills	Credit Hours: 3
This course provides a student by basic skills for using computer at studying environment, library, and at home. It presents the knowledge of basic computer and information technology concepts. The course provides the knowledge needed to operate and utilize the operating system and office software package, and to use the computer for Internet access and electronic communication.	
Course Code BEC01 Course Name Mathematics	Credit Hours: 3

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This course is designed to provide you with a solid foundation in the fundamentals of mathematics. The topics included in Basics of Mathematics are: Fundamentals of Algebra (Powers and Roots) Arithmetic (the basic operations of addition, subtraction, multiplication, and division), Types of Functions, Algebra (the use of symbols and letters to represent numbers and solve equations), Trigonometry (the study of trigonometric ratios and relationships between side lengths and angles of triangles), Calculus (the study of rates of change and accumulation), Limits and Continuous.		
Course Code BCO01 Course Name Computer Fundamentals		Credit Hours:3
This course will introduce: 1) fundamental electronic data processing concepts and associated terminologies. 2) the development of computers and computer application. 3) the impact of computer on society. Furthermore, peripherals of an actual computing system CPU configuration. Device interfaces. Binary number systems, and professional ethical issues in computing will be discussed.		
Course Code BCO02 Course Name Computer Programming I		Credit Hours: 3
This course will cover the abstract and structure of programming, problem-solving techniques and tools, flowcharts and algorithms. Abstract data types and their list, variables declarations and memory locations. Arithmetic operators, operator's precedence, equality and relational operators, cohesion and coupling. Abstract programming structure types: sequences, selection (decision making), iteration, multiple iteration, and errors types and Program structure correctness and verification, one and multi-Dimensions arrays (basics using and operations).		
Course Code BUST05 Course Name Islamic Culture		Credit Hours:3
يعد مقرر الثقافة الإسلامية من أهم المقررات الدراسية وذلك للدور الذي يساهم فيه هذا المقرر في تكوين الشخصية المسلمة القادرة على قراءة ماضيها وفهم واقعها والإسهام الإيجابي في بناء المستقبل بناءً يوافق شرع الله ويلبي احتياجات العصر من غير إفراط ولا تفريط (بوسطية). ويتم أخذ هذا المقرر بطريقة الحوار والمناقشة، كما يكلف فيه الطلبة بالبحوث التي تخدم أهداف المقرر. ويتم تقييم الطلبة عن طريق المشاركة الصفية واللاصفية والأعمال التي يكلف بها الطلبة والاختبارات النصفية والنهائية.		
Course Code BEC03 Course Name Technical English		Credit Hours: 4
The current course provides students with the language basics of everyday English to help them communicate in different real-life situations. The course focuses more on real life conversations and the basic grammars that will help each one in his major. Moreover, the course provides students with plenty of writing and speaking practices. The updates of the book allow students to be posted with the latest language uses and functions that are easy and useable in real life situations.		
Course Code BUST03 Course Name Life Skills		Credit Hours: 2
يهدف مهارات الاتصال إلى إلمام الطالب بالنظريات والمفاهيم الأساسية في مجال الاتصال الإنساني وتنمية المهارات الأساسية في مجال التواصل الجيد مع الذات ومع الآخرين، وتعزيز ممارستها في حياته اليومية والعملية باستخدام أساليب جديّة تعتمد على التدريب والتقويم المتنوع والفعال، بما يساهم في إنجاح حياته الخاصة والعملية على حد سواء.		

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Course Code BCO03 Course Name Computer Programming II	Credit Hours:3
This course will cover the rest of structure programming concepts and advance topics such as arrays, functions, pointers, strings, structures, files, and introduction to object-oriented programming.	
Course Code BIS401 Course Name Fundamentals of Information Systems	Credit Hours: 3
This is a foundation course that introduces students to the definitions, concepts and tools used in the IS field. It focuses on discussing the business implications of information systems. Topics covered include Business Information Systems, Database Systems and Applications, Telecommunications and Networks, Electronic and Mobile Commerce and Enterprise Systems, and Decision Support Systems.	

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Second Year Courses

Course Code BIT205 Course Name	Credit Hours: 3
Course Code BIT408 Course Name Database administration	Credit Hours: 3
This course provides the comprehensive knowledge about relational database management system in administrative approach to integrate in enterprise level of database in network environment which encompasses with oracle database instances Management, database installment in network environment, implementing user role and privileges, multitenant database management, back and recovery.	
Course Code BCO06 Course Name Database Systems and Design	Credit Hours: 3
This course covers the fundamental concepts of database systems. Topics include data models (ER, relational, and others); query languages (relational algebra, SQL, and others); implementation techniques of database management systems (index structures, concurrency control, recovery, and query processing); management of semi structured and complex data; distributed and NoSQL databases.	
Course Code BIS203 Course Name Linear Algebra	Credit Hours: 3
This course is designed to provide you with a solid foundation in the fundamentals of mathematics. It gives the students mathematical tools for solving engineering problems. This course includes the study of Determinants - Matrices - Systems of Linear Algebraic Equations. Complex Numbers and Functions of Complex Variables, Vector Calculus.	
Course Code BCO04 Course Name Data Structures and Algorithms	Credit Hours: 3
This course focuses on the design and implementation of computer programs in a high-level language, with emphasis on proper design principles and advanced programming concepts, including dynamic data structures and recursion. Efficient design, implementation and debugging techniques are stressed. Basic concepts of data structures such as strings, lists, arrays, stacks, queues, trees, graphs and analysis and design of efficient algorithms for searching, sorting and merging are examined	
Course Code BCO07 Course Name Object-Oriented Programming	Credit Hours: 3

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This course let students practice writing programs for Object Oriented programming (OOP) using Java Programming language. It offers students opportunity to learn how to solve problems and build applications using object-oriented techniques. It includes an introduction to object-oriented design, history and advantages of object-oriented design, introduction to object-oriented programming concepts, classes, objects, data encapsulation, constructors, access modifiers, const vs non-const functions, static data members & functions, function overloading, composition, aggregation, inheritance, polymorphism, abstract classes and interfaces, and exception handling.

Course Code BCO05 Course Name Introduction to Web Design	Credit Hours: 3
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Discover the basics of web design using HTML and CSS. No prior knowledge of HTML or web design is required. Throughout the course you will get the info you need to plan and design effective web pages. Come away knowing how to implement web pages by writing HTML and CSS code. And discover ways to enhance web pages with the use of page layout techniques, text formatting, graphics, and images

Course Code BIS408 Course Name Software Engineering	Credit Hours: 3
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The Software Engineering course provides an in-depth exploration of the principles, methodologies, and practices involved in software development. This course covers the entire software development life cycle (SDLC), including requirements gathering, design, implementation, testing, deployment, and maintenance. Students will engage in hands-on projects that utilize modern development tools and techniques, fostering collaboration and problem-solving skills. By the end of the course, participants will have the knowledge and experience necessary to design, develop, and manage software projects effectively.

Course Code BCO03 Course Name Probability and Statistics	Credit Hours: 2
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This course in statistics will provide students with the knowledge of how to display data, how to calculate central tendency and scattering measurement. The course presents an introduction to probability, an explanation of the concept of random variables and the probability distributions (discrete and continuous), and kinds of correlation and regression. It also explains how to use SPSS to deal with statistical subjects.

Course Code BIT405 Course Name System Analysis and Design	Credit Hours: 3
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To introduce students to the relative complexity of information requirements, systems analysis and design within a business organization, and to introduce students to the concepts, formal techniques, tools and methods used in the analysis, design and implementation of information systems. The course approaches the development of information systems from a problem-solving perspective. This course builds upon concepts to which the student has been exposed to in previous classes.

Course Code BIS406 Course Name Financial Accounting	Credit Hours: 3
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This course is an introduction to the fundamental concepts of financial accounting as prescribed by U. S. generally accepted accounting principles (GAAP) as applied to transactions and events that affect business organizations. Students will examine the procedures and systems to accumulate, analyze, measure, and record financial transactions. Students will use recorded financial information

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to prepare a balance sheet, income statement, statement of cash flows, and statement of shareholders' equity to communicate the business entity's results of operations and financial position to users of financial information who are external to the company. Students will study the nature of assets, liabilities, and owners' equity while learning to use reported financial information for purposes of making decisions about the company. Students will be exposed to International Financial Reporting Standards (IFRS).

Course Code BC08 Course Name Operating Systems

Credit Hours: 3

Introduction to the fundamental principles of operating system design. The concepts and algorithms covered in the course are based on those used in both commercial and open-source operating systems. We present a large number of examples that pertain to the most popular and the most innovative operating systems. We also assign some simple coding labs to help students understand important knowledge and representative algorithms used in operating systems, such as CPU scheduling, synchronization and virtual memory.

Course Code BIS411 Course Name E-Business

Credit Hours: 3

This course describes the basic principles of e-business technologies. Upon the completion of this course, students should have a good working knowledge of e-business concepts, applications and technologies (e.g. e-business infrastructure, technology required for e-business, e-business marketplace, e-Commerce, B2B e-business, e-business strategy, e-procurement, cu

Third Year Courses

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Course Code BIT404 Course Name Data Communication and Networks	Credit Hours: 3
Technologies related to data communication and networking may be the fastest growing in our culture today. People use the Internet more and more every day. They use the Internet for research, shopping, airline reservations, checking the latest news and weather, and so on. This course is given to provide students with fundamentals of computer networking, the OSI model, and the TCP/IP model. A discussion of the physical layer of the Internet model and the transmission media is given. Students will learn switching techniques which can be used in several layers. A discussion of the data-link layer and network layer of the Internet model will also be given.	
Course Code BIS403 Course Name Fundamentals of Cybersecurity	Credit Hours: 3
This course provides an introduction to cybersecurity. Topics include hacking, social networks, privacy, cryptography, legal aspects, social implications, password management, digital forensics, computer networking, wireless security, and ethical issues. The course focuses on individual users and their role in protecting themselves from various cybersecurity threats. No technical experience needed.	
Course Code BIS412 Course Name Ethics of IS	Credit Hours: 2
This course considers the ethical issues that arise as a result of the increasing use of computers, and explores and discusses key ethical, legal and professional issues and responsibilities in computing and other related fields. It examines emergent technologies within frameworks that highlight their ethical, legal and social implications. Topics include privacy, confidentiality, security, intellectual property, software piracy, cybercrime, digital identity, software reliability, risk, safety, and professional standards of conduct and codes of ethics.	
Course Code BIT407 Course Name IS Project Management	Credit Hours: 3
An IS Project Management course syllabus typically covers foundational project management concepts, IS-specific project management methodologies, and tools for managing IS projects effectively. Key topics include project initiation, planning, execution, monitoring & controlling, and closure, along with an introduction to agile and traditional project management approaches. Students will learn to manage project scope, time, cost, quality, resources, communications, and risks, often culminating in a project where these concepts are applied	
Course Code BIT405 Course Name Business Intelligence	Credit Hours: 3
This course aims to provide students with understanding of business intelligence and analytics and its role in developing and sustaining competitive advantage for business organizations. This course will equip students with the necessary knowledge to apply business intelligence and analytics in various business contexts and learn skills required to scientifically and creatively deal with data in order to assist business organizations in enhancing their competitive edge.	
Course Code BIS415 Course Decision Support Systems	Credit Hours: 3

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The main objective of this course to develop students the required skills and gives them knowledge of the various decision-making models so that decisions be based on logical and mathematical foundations under different circumstances such as in cases of uncertainty, lack of information or certainty. It equips students with a mathematical framework on which a set of statistical algorithms is built to help the decision-makers. It acquaints the students with a variety of decision-making theories such as (the Decision Theory itself, Pragmatic Theory and Players Theory) that can be used in various applications.		
Course Code BIS416 Course Name Principles of Marketing		Credit Hours: 3
Principles of Marketing provides students with a comprehensive understanding of the marketing discipline. The purpose of this course is to provide you with a fundamental understanding of key marketing concepts and principles. Such an understanding will facilitate a greater appreciation of the value of marketing processes and activities for organizations (both profit and non-profit), destinations, ideas and individuals.		
Course Code BIS417 Course Name Enterprise Resources Planning		Credit Hours: 3
Explains and demonstrates how Enterprise Resource Planning (ERP) Systems, a group of integrated software modules, run virtually all business processes in an organization. ERP Systems include modules such as management accounting, sales, logistics, production/materials management, procurement, and human resources.		
Course Code BEC04 Course Name Research Methodologies		Credit Hours: 2
يعد هذا المقرر من أهم المقررات الدراسية كونه يمكن الطالب الجامعي من المهارات التي تساعده على إنجاز بحث التخرج وهو يعمل على إكسابه مجموعه من المعلومات والمعارف حول العلم والمعرفة والبحث العلمي ومناهجه وينمي الاتجاهات الإيجابية لدى المتعلم تجاه البحث العلمي ويتحقق ذلك من خلال مجموعه من الأنشطة الفاعلة (طرائق وأساليب تدريس، ووسائل) كطريقة الحوار والمناقشة والتعلم الذاتي والبحث والتطبيقات مع استخدام الباوربونت عن العرض وتقييم الطالب من خلال المشاركة الصفية والتكليفات المصاحبة ، الصفية واللاصفية ، والاختبار النصفى والنهائى.		
Course Code BIS420 Course Name Artificial Intelligence		Credit Hours: 3
This course introduces the basic principles in artificial intelligence. It covers simple representation schemes, problem solving paradigms, constraint propagation, and search strategies. Areas of application such as knowledge representation, natural language processing, expert systems, vision and robotics are explored. The Python programming language is also introduced.		
Course Code BEC07 Course Name Field Training		Credit Hours: 3
This course offers instruction on using computer tools and their applications in a variety of industries. training in institutions in the public or private sectors, under the direction of department faculty. that		

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the students' acquisition of information and experience occurs through field training, which is supervised by real-world situations. exercising in the training program's academic component. Students benefit from field experience, which also contributes to on-the-job training. There should be opportunities for non-learning during field training. present in classrooms.

Course Code BIS413 Course Name Mobile Computing

Credit Hours: 3

This course studies basics and emerging topics in mobile computing, including basic communication theory of wireless mobile networks as well as the software development for mobile devices. Newly-emerging areas of mobile computing involving machine intelligence and Bot will also be studied in depth. Basics of mathematical theories and algorithms for communications will be introduced for various systems such as Internet, mobile cellular networks, wireless LAN, GPS global positioning systems. Augmented reality and QR code applications will be introduced. Software development tools for mobile apps and Bot technologies in mobile devices such as smartphones and tablets will be introduced as project-based learning.

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Fourth Year Courses

Course Code BIS418 Course Name Advanced Web Application Development	Credit Hours: 3
This course reviews basic concepts and techniques for developing applications within computer network environment. Emphasis is placed upon the knowledge and skills to develop web-based applications on both client and server side especially in Internet environment. It also includes the design and creation of interactive web sites that provide access to databases. Other topics including Cascading Style Sheets, XML, AJAX and Web Services are also introduced. This course requires Introduction of Web Design, Introduction of Database as a pre-requisite. The lectures of this course will be taught through lectures, projects, exams, presentations and assignments.	
Course Code BIS410 Course Name Cloud Computing	Credit Hours: 3
The course presents a top-down view of cloud computing, from applications and administration to programming and infrastructure. Its main focus is on parallel programming techniques for cloud computing and large-scale distributed systems which form the cloud infrastructure. The topics include: overview of cloud computing, cloud systems, parallel processing in the cloud, distributed storage systems, virtualization, security in the cloud, and multicore operating systems. Students will study state-of-the-art solutions for cloud computing developed by Google, Amazon, Microsoft, Yahoo, VMWare, etc. Students will also apply what they learn in one programming assignment and one project executed over Amazon Web Services.	
Course Code BCO05,06 Course Name Graduation Project I, II	Credit Hours: 3
This course focuses on the graduation project design stages. It provides students with the opportunity to engage in an activity that will allow them to demonstrate their ability to apply the knowledge and skills they have gained throughout their years in the educational system. At the end of this course, students must deliver a project with a major component that has passed through the design, analysis, implementation, testing, and evaluation stages.	
Course Code BIS402 Course Name Supply Chain Management	Credit Hours: 3
This course teaches concepts useful in efficiently managing Supply Chains. Topics covered include: the role of Supply Chain Management in overall competitive strategy, terms, definitions, Supply Chain examples, key performance measures, and tools for improving Supply Chain performance. The level of discussion varies from long-term strategic planning to daily control of Supply Chain & business processes.	
Course Code BIS422 Course Name Internet of Things	Credit Hours: 3
This course is designed to introduce learners to the use, benefits and impact of IoT not only in industry and business but in our day-to-day lives. Students will learn how data is communicated, collected, analyzed and abstracted, and delivered to the end user to make life safer, more efficient	

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and to assist in achieving better data to make better decisions. The IoT Application Framework will be introduced as a practical application development and deployment roadmap to assist the end user in successfully implementing IoT solutions. Students will also learn some of the challenges inherent in setting up and using IoT effectively including cybersecurity concerns and privacy issues.

Course Code BIS407 Course Name Enterprise Architecture and Systems

Credit Hours: 3

This Course presents the basic concepts and methodologies for the discipline known as Enterprise Architecture within a framework, structure, and methodology. Per Gartner, “Enterprise architecture is the process of translating business vision and strategy into effective enterprise change by creating, communicating and improving the key requirements, principles and models that describe the enterprise's future state and enable its evolution. The scope of the enterprise architecture includes the people, processes, information and technology of the enterprise, and their relationships to one another and to the external environment. Enterprise architects compose holistic solutions that address the business challenges of the enterprise and support the governance needed to implement them

Course Code BIS407 Course Name Human Resources Management

Credit Hours: 3

This course is an introduction to the human resource management (HRM) function and related elements and activities to examine the role of the human resource professional as a strategic partner in managing today's organizations. Key functions such as recruitment, selection, development, appraisal, retention, compensation, and labor relations are examined. Implications of legal and global environments are appraised and current issues such as diversity training, and the rising cost of benefits are analyzed. Emphasis is placed on the modern day importance of HRM at the corporate level as well as the view of HRM from the perception of both management and subordinate employees.

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